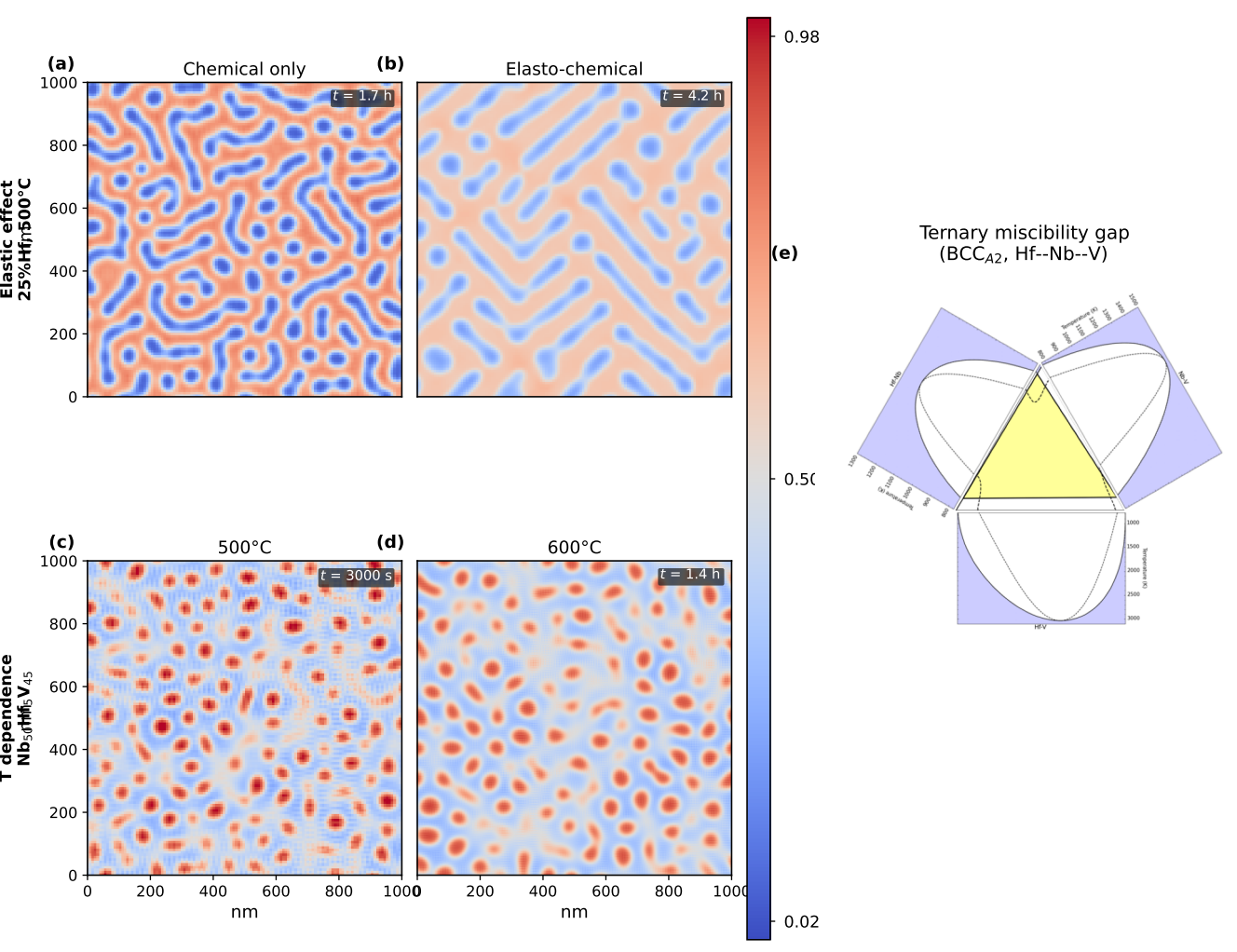

(a)
Chemical only
(b)
Elasto-chemical
t = 1.7 h
t = 4.2 h
Elastic effect
25%Hf, 500°C
(c)
500°C
(d)
600°C
t = 3000 s
t = 1.4 h
T dependence
Nb50Hf5V45
0
200
400
600
800
1000
nm
0.98
0.50
0.02
(e)
Ternary miscibility gap
(BCC_A2, Hf--Nb--V)
Hf-Nb
Hf-V
Nb-V
Hf-V